

Applied Mathematics-I B. Tech. I (Semester I) Computer Science and Engineering

Note: Question 1. is compulsory. Attempt any two questions from question 2, 3 and 4.

Time: 1 hour

Max. Marks: 15

1. (a) Evaluate $\int_0^{\infty} e^{-x^2} dx$.
(b) Evaluate $\int_0^{\pi/2} \sin^2 \theta \cos^5 \theta d\theta$.
(c) Define improper integral of first kind with example.
(d) Find polar form of $\frac{-1+\sqrt{3}i}{2}$.
(e) Show that $f(z) = [z]^2$ is analytic function at $z = 0$ and nowhere else. ($1 \times 5 = 5$)
2. Solve $x^4 - x^3 + x^2 - x + 1 = 0$. 5
3. Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. 5
4. (i) Discuss convergence of $\int_1^{\infty} \frac{dx}{(x-1)^p}, (p > 0)$. 2.5
(ii) Expand $\cos^3 \theta \sin^2 \theta$. 2.5

B.Tech-1st Semester (Applied Mathematics)

Max.Marks:15

Time allowed: 1hr

Note: Section A is compulsory and attempt any two questions from section B. Mention your group on the top of the answer sheet.

SECTION - A

1. (a) Check whether $f(z) = |z|^2$ is analytic?
- (b) Define harmonic functions.
- (c) Find fourth roots of unity.
- (d) Evaluate $\int_0^{\pi/2} \sqrt{\tan x} \, dx$
- (e) Show that the improper integral converges $\int_1^{\infty} \frac{dx}{x^n}$ converges when $n > 1$. (5 * 1 = 5)

SECTION - B

2. (a) Prove that $r(1/2) = \sqrt{\pi}$.
- (b) Discuss the convergence or divergence of $\int_{-\infty}^{\infty} \frac{dx}{e^x + e^{-x}}$
3. If $f(z) = u + iv$ is analytic function of $z = x + iy$ and $u - v = e^{2xy} \cos(x^2 - y^2) + e^{-2xy} \sin(x^2 - y^2)$ then find u, v and the analytic function $f(z)$.
4. (a) Expand $\cos^4 \theta \sin^3 \theta$ in terms of $\sin n\theta, \cos n\theta$.
- (b) Evaluate the integral $\int_0^3 \frac{dx}{x^2 - 3x + 2}$.

SUBJECT: APPLIED MATHEMATICS I (B Tech. CSE)

Time Allowed: 1hr

Note: Section A is compulsory and attempt any two questions from section B.

Max. Marks: 15

SECTION - A

1. (a) Discuss convergence of $\int_0^{\infty} \frac{dx}{e^x + e^{-x}}$.
- (b) Show that $\int_0^{\pi/2} \sin^2 \theta \cos^4 \theta d\theta = \frac{\pi}{32}$.
- (c) Evaluate $\int_0^{\infty} e^{-x^3} dx$.
- (d) Find the values of $\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{3/2}$.
- (e) Show that $f(z) = \bar{z}$ is not differentiable.

(5 × 1 = 5)

SECTION - B

2. Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.
3. (i) Prove that $\sin^4 \theta \cos^2 \theta = \frac{1}{2^5} (\cos 6\theta - 2 \cos 4\theta - 2 \cos 2\theta + 2)$.
(ii) Find roots of $x^4 - x^3 + x^2 - x + 1 = 0$.
4. (i) If $f(z) = u + iv$ is an analytic function, find $f(z)$ if $u(x, y) = 2x + y^3 - 3x^2y$.
(ii) Determine a, b, c, d so that the function $f(z) = (x^2 + axy + by^2) + i(cx^2 + dxy + y^2)$ is analytic.
(2 × 5 = 10)